College of the Siskiyous Annual Program Update

Program: BIOLOGY AND ENVIRONMENTAL SCIENCES FOR AY 2017/18 Date: December 11, 2018

Faculty Completing Form: VARTY AND CLARKE

1. Describe annual progress / changes / events in the program.

We think of our department as having three different components, our pre-health courses, general education courses, and majors courses and degrees. Thus, we have split this analysis up into those three groups.

Pre-Health

- We lost half of our adjunct faculty in Biology over the past year. As a result of this loss, we were forced to decrease our offerings in Microbiology, Human Anatomy (in addition to not offering Human Biology, which is a GE class, at all).
- Consequently, we will have a new full-time faculty member in Biology starting spring 2019. We thank our dean for perceiving this need and support our CQIP to fund this position.
- With our new faculty member on board for all of AY 19-20, we anticipate increasing our offerings of Anatomy and Microbiology compared to what they are this year (see CQIP for increased instructional supplies for AY 19-20). This is part of the reason we believe our total departmental offerings will increase by 9 courses in AY 19-20 (See Appendix 1, our proposed AY 19-20 schedule). Another component of this projected increase is more General Education courses in order to meet student demand (see discussion of this below).
- Dave Clarke will be on sabbatical leave for the spring 2019 semester to develop online anatomy and physiology.

General Education

- The number of GE courses offered in AY 15-16, 16-17, 17-18 and 18-19 are listed in the table below:

	AY 15-16	AY 16-17	AY 17-18	AY 18-19
Introduction to Life Science	1	1	3	4
Environmental Principles	0	0	3	4
Human Biology	2	2	1	0
Field Botany	0	1	1	1
Environmental Science	1	2	2	2
Totals	4	6	10	11

- We think this growth (nearly a three-fold increase in GE course offerings in 4 years) is impressive and indicates strong and growing student interest in our courses. We feel we could continue this growth in our GE offerings by making some additions to our schedule in the future (These additions are reflected in our proposed schedule for AY 19-20):
    - We would like to offer an online GE laboratory course (Environmental Science). The lack of a fully online laboratory course is currently a barrier to COS students receiving a degree online. Varty has agreed to develop it over the summer of 2019 and work towards getting it approved for offering in the CCC online course exchange. Ideally, she could offer it for the first time in Fall of 2019.
    - We continue to experience a high demand for Field Botany. This course is typically one of our first GE options to fill and is the one Varty gets the most inquiries about from current and prospective COS students. We believe there would be high interest in the community for an additional section of this course and our botany short courses: Trees and Shrubs and Wildflower Identification. All three of these courses include field trips. This opportunity for growth is the reason we are asking for additional funds for mileage and funds to support van drivers.
- We are also concerned about the strain this growth puts on our laboratory technician. Mandy had a little break this year with two less Microbiology classes. However, that will increase again next academic (Note that none of

the growth we propose above in our GE offerings will add to her workload as both of the proposed courses are primarily online).

Majors

- We continue to offer one course geared toward science majors each semester. Enrollment in our courses (BIO 2100 and BIO 2200) has been stable.

Continued professional growth

- We continue to seek out opportunities to develop professionally, but find it increasingly difficult to do so on the FLEX budget we are allotted. Examples of our professional growth activities during AY17-18, some of which cost us a considerable amount of personal funds, include:
    - Varty completed the American Museum of Natural History Genetics, Genomics, and Genethics course.
    - Clarke attended the Human Anatomy and Physiology conference.
    - Varty attended the University of Michigan Biological Station summer lecture series.
    - Varty attended the California Native Plant Society Bryophyte workshop at the Siskiyous Field Institute
    - Varty published a paper in the Journal of Natural History Education and Experience
    - Varty and Clarke attended multiple Elumen trainings
    - Varty taught herself how to use Proctorio

2. <u>Describe progress on assessing SLOs and PLOs for the program.</u>

SLOS are being accessed in all biology and environmental science offerings. We have no 'program' slos, and so are not assessing any.

3. <u>Discuss any support or obstacles encountered by the program.</u>

- No department credit card for our Technician. Having a credit card would streamline our purchases and require less work by the business and instructional offices. She is extremely busy; this may free up some of her time to allow her to provide our students with more support.

- Delayed budgetary information. We are not notified of our budget until well after the semester starts, making prioritization decisions on what to purchase problematic
- Timing of instructional equipment requests. Non-instructional areas are allowed to request funds at any time of year, while we are limited to CQIPS submitted with program review.
- Increasingly, many laboratory activities require computer access. The ASC computers are seldom available. Computer classrooms are scheduled for other courses and have also not been available for our use.
- Funds for professional development are inadequate.
- This year we were not given a budget to have our microscopes services. It is important that these resources be maintained (see CQIP for details).
- The transitions in the ASC have caused issues for our Work Experience in Biology students, made it more difficult for us to coordinate tutor support for our students, and caused issues for us when it came to accessing computers. We would love to see the institution funnel more resources toward this resource for us and our students.

4. Describe program budgetary needs or implications. (Attach completed CQIP form.)

- Increased supplies funding reflecting two additional sections of Microbiology (see CQIP)
- Updated physiography software (see CQIP)
- 24 laptops for biology (see CQIP)
- New cadaver. We have been able to 'stretch' our cadaver dollar in the past by retaining the specimens for several years. The willed body program is now enforcing retention limitations. This will require us to annually acquire a new cadaver (and send back one, which also costs) (see CQIP).
- Reinstatement of our microscope repair budget (see CQIP).
- Increased funding for mileage and van drivers to grow our natural history offerings (see CQIP).

Appendix 1: Our model AY 18-19 schedule

COURSE #	TITLE	UNITS
	FALL 2019	
ENVS 1002	ENVIRONMENTAL SCIENCE (ONLINE?)	4
BIOL 0891	BIOPREP (ONLINE)	1
BIOL 1000	INTRODUCTION TO LIFE SCIENCES	4
BIOL 1000	INTRODUCTION TO LIFE SCIENCES	4
BIOL 1400	ENVIRONMENTAL PRINCIPLES (ONLINE)	3
BIOL 1400	ENVIRONMENTAL PRINCIPLES (ONLINE)	3
BIOL 1500	HUMAN BIOLOGY	3
BIOL 2200	GNERAL BIOLOGY I	5
BIOL 2600	MICROBIOLOGY	4
BIOL 2600	MICROBIOLOGY	4
BIOL 2700	HUMAN ANATOMY	4
BIOL 2700	HUMAN ANATOMY	4
BIOL 2700	HUMAN ANATOMY	4
BIOL 2800	HUMAN PHYSIOLOGY	5
	HUMAN PHYSIOLOGY - LAB	
BIO 1028	TREE AND SHRUB IDENTIFICATION	0.5
	SPRING 2020	
ENVS 1002	ENVIRONMENTAL SCIENCE	4
BIOL 0891	BIOPREP (ONLINE)	1
BIOL 1000-1	INTRODUCTION TO LIFE SCIENCES	4
BIOL 1000-2	INTRODUCTION TO LIFE SCIENCES	4
BIOL 1400	ENVIRONMENTAL PRINCIPLES (ONLINE)	3
BIOL 1400	ENVIRONMENTAL PRINCIPLES (ONLINE)	3
BIOL 1500	HUMAN BIOLOGY	3
BIOL 1700	FIELD BOTANY (HYBRID)	3
	FIELD BOTANY-FIELD TRIPS	
BIOL 1700	FIELD BOTANY (HYBRID)	3
	FIELD BOTANY-FIELD TRIPS	
BIOL 2200	GENERAL BIOLOGY II	5
BIOL 2600	MICROBIOLOGY	4
BIOL 2600	MICROBIOLOGY	4
BIOL 2700	HUMAN ANATOMY	4

BIOL 2700	HUMAN ANATOMY	4
BIOL 2700	HUMAN ANATOMY	4
BIOL 2800	HUMAN PHYSIOLOGY	5
	HUMAN PHYSIOLOGY - LAB	